[PR-AIE]

Day-2 - Python Basics - Feb25

* To uninstall a software:

→ control panel → Programs → uninstall a software

* Lec-4 - Installation done ✓

Lec-5 - Python features

→ ~~Features~~ Features of a Language are nothing but services or facilities provided by Language developers. in the languages, which are used by Language programmers, → in development of Real time applications.

* Ex: If u want to buy a product, you will go for the one with good facilities and services. (i.e mobile with good features)

→ We have 11 features in Python

1. Simple & easy to write code [Because of 3 factors]

→ Less lines & more meaning in code compared to other programming lang.

2. Platform independent language

a-1 → Why simple? Python provides "Rich set of modules". So that python programmer can Re-use the predefined code which is present in modules, without writing our own source code. (Continue in next page)

Lec-6 - Python features - part-2

→ Definition of module:

Function: A function in Python is a block of reusable code that performs a specific task.

→ Instead of writing the same code again & again, you define a function once and call it whenever needed.

→ Ex: import calendar as cl

print (cl.month(2025,4))

↳ month( ) is a function in calendar module

→ * Data members:

math.pi gives 3.1415 } These are data members.
math.e gives 2.718 }

Data members in a module are the variables defined at the module level that store data and can be accessed after importing the module.

→ A module can contains; variables (Data members), functions, Classes and constants.

In math.py file, we have

pi = 3.1415 } stored in variables with these names.
e = 2.718 }

In Python, a module is simply a .py file that contains variables, functions and classes.

→ Data members in a module are the variables defined inside that module. They store data that can be accessed after importing the module.

A module is a collection of functions, Data members and class names.

Factor 2:- * Python provides in-built facility called Garbage collector, to eliminate the unused memory space.

→ Garbage collector collects unused memory space and improves the performance of python based Applications

* Def of Garbage collector:- [Python Background program]

→ The GC in python is an automatic memory management system that uses reference counting & cyclic garbage collection to reclaim (free) memory occupied by objects that are no longer in use.

You don't need to manually delete memory like in C or C++.

→ * Collects un-used memory space & improves performance.

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* Factor-3: Python provides user-friendly syntaxes and helps programmer to write error-free programs in limited span of time.

→ Lec-7- Python features- part 3

1. Why python is Platform independent?

→ In software development, we have 2 types of languages:

→ 1. Platform Dependent Language [C, C++]

2. Platform Independent language. [Python, Java]

→ 1. Platform dependent language: [C, C++]

Here, Platform means the Environment where a program runs.

It is a combination of hardware & OS.

Ex [Windows + Intel CPU ; Linux + ARM processor]

Simply the type of OS being used to run our program.

OS: It is system software that manages computer hardware & provides services for application programs

It manages all hardware & software resources of a computer

Think of OS as like this;

Hardware = Body parts

OS = Brain & nervous system

Applications = Skills (talking, writing, running)

without the brain (OS), body parts are useless.

OS is a software and it acts as Resource Allocation Manager and Resource deallocation manager.

(or)

OS is one of interface b/w Program and computer Hardware (memory, processor, I/O Devices etc).

→ C, C++ Programming Lang Environment;

Data Types; used for allocating memory space ~~and~~ for Storing inputs.

→ A datatype tells the computer what kind of data a variable can store and how much memory is needed for that data.

Datatype specifies the type of value a variable can hold and the operations that can be performed on it.

~~ex: int, float, character, float, string, bool, double, unsigned int~~

Ex: int, float, double, char, Array, pointer, structure, unsigned int

OS;	int	float	double	char
DOS;	2 bytes	4 bytes	8 byte	1 byte
UNIX;	4 bytes	8 bytes	16 bytes	1 byte

} differs from one OS to another OS.

So, C is dependent on OS as datatypes memory varies from one OS to another OS.

* Since C, C++ Languages Related data types takes different memory spaces on different OSes, so these are platform dependent.

If we write a C program in UNIX based OS, it must be run/works only for UNIX based OS, & doesn't work on DOS based OS.

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Data types in Python; list, tuple, int, float, bool etc, str, dict

→ All values are stored in objects & all data types takes same memory space in every OS.

* No memory restrictions in objects in python.

→ Every datatype is stored here in the form of object.

→

Java Programming Lang Environment

Java Data Types;

Data Type	DOS	UNIX
byte	1	1
short	2	2
int	4	4
long	8	8
float	4	4
double	8	8
boolean	0	0
char	2	2

The above Datatypes are called Pre-defined Data Types

→ Java Predefined datatypes takes same memory space on all types of OSes and hence Java lang is platform independent language.

The limitations of Java lang are

(i) The predefined datatypes allows us to store limited memory space sized value. Here datatype cannot store if it exceeds its limit.

It has memory space Restriction

In Java;	In Python;
int a=9; → works fine	a=9 ✓
a is integer data type.	a=99 ✓
→ When we update a to 99999999999	a=99999999999 ✓
a=99999999999, it gives error as integer number too large.	No Error Here

but python can store these big numbers also.

	In Python
(ii) jshell >int x=10, "Rajesh", 34.126	>>> s= 100, "as", 31.12
Error,	Print(s)
<identifier> expected	
int x=10, "Ravi", 34.56;	(100, "as", 31.12) ✓ No Error Here

The pre-defined Data types allows us store single value and unable to store multiple values of different type.

* This approach is called Non-object approach where we can store a single value only.

Here no objects are created.

The non-object approach in Java means writing programs without creating objects by using static methods and static variables.

→ In Java, if I want to store multiple values in single variables, we use OBJECT approach.

* Python is fully object approached based.

* Python objects allows to store multiple values where as Java objects have memory restrictions.

* In Java, when we define a Class & create 3 variables, then we can store only 3 values for that objected, but in python we can store unlimited values for a object.

*** In Python, all values are stored in the form of objects.

⟷ *** 'int' is a class in python.

* 'True' is a boolean class.

* Platform Independent Lang:

→ In this, Data types memory space remains same on all type OSs

→ In Effective platform independent lang, all types of values will store in the form of objects & they can store unlimited amount of data.

→ Java objects contains size Restricted where Python objects contains un-limited size and unlimited values can store.

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Lec-8- Python features - Part-3

→ Python is Dynamically typed Programming language.

→ In IT, we have 2 types of Programming language

1. static Typed PL. [C, C++, Java, C#.net, ---]
2. Dynamically " PL [Python]

→ 1. static Typed PL :-

In C, to store 100 in memory, ---

with help of this

(1) allocate memory space --- Data Type (int)

(2) Give name to memory space --- say - a

int a ----> Variable declaration --- mandatory in static typed PL.

→ Task; compute sum of 2 numbers

Take 10 & 20 & store in a & b

Compute a+b & store in c;

In C, C++

int a=10; → mandatory to write datatype

int b=20; → variable declarations

int c=a+b;

If u don't declare datatype, then memory will not be allocated & gives error as

Error:

cannot find symbol

Symbol: variable c

c=a+b

^^^

→ In static typed PL, it is mandatory for the programmer to specify variable declaration (Data type + identifiers), Else we get errors

→ Called Compile time Errors.

11.25 → double

11.25f → float

x=11.25, Here x is double

x=11.25f, Here x is float.

* The problem of static typed PL is that the programmer may not know the data type of value accurately.

* Ex: Date 22-04-2025; what is this in C?

How to declare it?

→ Here it stores only the declared data type values only but not allows other types of values

Ex: int i=10; ✓

i=20 ✓

i=20.10 ✗ ← Not stores; Cannot be updated.

i can be updated to only int values.

→

2. Dynamically Typed PL:-

Here, we need not to write variable declarations.

Internally Programming Language execution Environment automatically will decide the data type of value, which is Entered by us.

a=45 ✓

a="str" ✓ allowed in Dynamic typed PL

a=11.25 ✓

* Even if we don't know datatype, no problem here.

*

a=10

b=12.5

c=a+b

Print(c) → 12.5 ← float ✓

* The main advantage is

(i) Programmer Need not write Datatype

(ii) Depends on type of value, Execution Environment will automatically assign the data type.

*